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COLLAPSIBLE CARTS

Abstract

Embodiments of the present disclosure may include a collapsible cart configured to transition from a closed condition where it may be folded up to an open condition where it may be expanded for use, the collapsible cart including a rigid frame forming a compartment, the rigid frame having a front wall, a rear wall, a right sidewall, a left sidewall, and a bottom wall, the right sidewall and the left sidewall may be configured to fold inwardly in the closed condition. In some embodiments, the right sidewall including a first right panel rotatably coupled to a second right panel. In some embodiments, a retractable handle mechanism is disposed at, within or adjacent the back wall. The retractable handle mechanism includes a hand grip attached to a telescoping assembly. The telescoping assembly is pivotably coupled at a proximal end to the bottom of the rear wall.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 19/032,431, filed on Jan. 20, 2025 and issued as U.S. Pat. No. 12,304,546 on May 20, 2025, which in turn is: (1) a continuation-in-part of U.S. patent application Ser. No. 18/161,677, filed on Jan. 30, 2023, which is a continuation-in-part of U.S. patent application Ser. No. 17/712,032, filed on Apr. 1, 2022 and issued as U.S. Pat. No. 11,565,735 on Jan. 31, 2023, which is a continuation of U.S. patent application Ser. No. 17/143,116, filed on Jan. 6, 2021 and issued as U.S. Pat. No. 11,338,835 on May 24, 2022, which claims the benefit of the filing date of U.S. Provisional Patent Application Ser. No. 62/995,375 filed on Jan. 27, 2020 and also claims benefit of U.S. Provisional Patent Application Ser. No. 62/974,956 filed on Jan. 6, 2020, the disclosure of each is incorporated herein by reference in its entirety; and (2) a continuation of U.S. patent application Ser. No. 18/759,664, filed on Jun. 28, 2024, which is a continuation of U.S. patent application Ser. No. 18/542,495, filed on Dec. 15, 2023 and issued as U.S. Pat. No. 12,103,576 on Oct. 1, 2024, which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/576,750, filed on Mar. 6, 2023, the disclosure of which is incorporated herein by reference in its entirety, and also claims benefit of U.S. Provisional Patent Application Ser. No. 63/577,068, filed on Mar. 28, 2023, the disclosure of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] This patent document relates to collapsible carts, and more particularly, to collapsible carts for high load capacity that are capable of folding up for convenient storage.

DESCRIPTION OF THE RELATED ART

[0003] Collapsible carts have been widely used in household and commercial settings. They provide a convenient way to transport objects when needed, but can be folded or collapsed for storage.

[0004] The following patents and published applications are representative prior art: 1. U.S. Pat. No. 2,564,939 issued to Louis S. Weast on Aug. 21, 1951 for “Foldable Shopping Cart”; 2. U.S. Pat. No. 2,786,692 issued to Anne Jackson Timpson on Mar. 26, 1957 for “Collapsible Cart”; 3. U.S. Pat. No. 3,092,395 issued to Sol Mitty et al. on Jun. 4, 1963 for “Corrugated Shopping Cart and Parts”; 4. U.S. Pat. No. 3,135,527 issued to Philip B. Knapp on Jun. 2, 1964 for “Wheeled Market Carts”; 5. U.S. Pat. No. Des. 292, 135 issued to John W. Grube et al. on Sep. 29, 1987 for “Collapsible Cart”; 6. U.S. Pat. No. 4,765,644 issued to Laurence G. Bell on Aug. 23, 1988 for “Foldable Cart”; 7. U.S. Pat. No. 4,765,646 issued to Karen Cheng on Aug. 23, 1988 for “Collapsible Shopping Cart”; 8. U.S. Pat. No. 5,197,754 issued to Lyla B. Ward on Mar. 30, 1993 for “Collapsible Beach Cart”; 9. U.S. Pat. No. 5,244,219 issued to Sidney R. Hadlum on Sep. 14,

1993 for “Hand Held Carrier”; 10. U.S. Pat. No. 5,988,671 issued to Kevin G. Abelbeck et al. on Nov. 23, 1999 for “Collapsible Cart”; 11. United States Published Patent Application No. 2002/0050429 to Cory O. Nykoluk et al. on May 2, 2002 for “Pivotal Handle For Towable Baggage”; 12. United States Published Patent Application No. 2002/0139628 to Wen-Cheng Chang on Oct. 3, 2002 for “Retractable Handle Assembly”; 13. United States Published Patent Application No. 2002/0144874 to Cory O. Nykoluk et al. on Oct. 10, 2002 for “Pivotal Handle For Towable Baggage”; 14. U.S. Pat. No. 6,598,898 issued to Yong S. Chu on Jul. 29, 2003 for “Folding Cart”; 15. U.S. Pat. No. D477,916 issued to Cory O. Nykoluk on Aug. 5, 2003 for “Towing Member For A Piece of Baggage”; 16. U.S. Pat. No. 6,651,791 issued to Cory O. Nykoluk et al. on Nov. 25, 2003 for “Pivotal Handle For Towable Baggage”; 17. United States Published Patent Application No. 2004/0211635 to Chen-Tien Lu on Oct. 28, 2004 for “Apparatus for Mounting Telescopic Handle on Trunk”; 18. U.S. Pat. No. 6,918,474 issued to Cory O. Nykoluk on Jul. 19, 2005 for “Towable Wheeled-Backpack”; 19. United States Published Patent Application No. 2009/0212536 to Maria I. Tadeo on Aug. 27, 2009 for “Collapsible Rolling Tote Bag”; 20. U.S. Pat. No. 7,731,221 issued to Suzan L. Bess on Jun. 8, 2010 for “Collapsible and Portable Wheeled Dolly Particularly Suitable for Use by Students or Others in the Transport of Items”; 21. U.S. Pat. No. 7,066,476 issued to the named inventor Richard Elden on Jun. 27, 2006 for “Side Attachable Cover/Seat for a Cart Carrying Box”; 22. U.S. Pat. No. 7,147,243 issued to Darren Kady on Dec. 12, 2006 for “Accessories for a Collapsible Rolling Caddy”; and 23. U.S. Pat. No. 8,439,374 issued to the named inventor Richard Elden on May 14, 2013 for “Lightweight High Load Capacity Folding Utility Cart with Unique Support Structure and Ergonomic Handle.”

[0005] Because of the collapsible nature of the prior art cart design, the sidewalls may not be sufficiently sturdy to allow for transporting heavy objects. None of the prior art has effectively addressed this problem. Thus, the need for improvements to collapsible carts still remains.

SUMMARY

[0006] Stackable collapsible carts are disclosed herein. Embodiments of the present disclosure may include a collapsible cart configured to transition from a closed condition where it may be folded up to an open condition where it may be expanded for use. The collapsible cart may include a rigid frame forming a compartment. The rigid frame having a front wall, a rear wall, a right sidewall, a left sidewall, and a bottom wall. The right sidewall and the left sidewall may be configured to fold inwardly in the closed condition.

[0007] In some embodiments, the right sidewall includes a first right panel rotatably coupled to a second right panel. In some embodiments, the second right panel proportioned to fit within an opening in the first right panel. Embodiments may also include a first lock assembly comprising a first track and a first slideable member. The first track formed along the first right panel and the second right panel, and extending from a first position on the first right panel to a second position on the second right panel. The first slideable member may be cooperatively engaged to the first track. The first slideable member may be movable along the first track between an open position to a closed position to selectively lock the first right panel to the second right panel. In some embodiments, the first slideable member may be in the open position when disposed along the first track adjacent the first position of the first track while not disposed along the second right panel and may be in the closed position when disposed along the first track adjacent the second position of the first track while being disposed across both the first right panel and second right panel.

[0008] In some embodiments, the second right panel may include a ribbed wall with a plurality of ribs. The ribbed wall may be disposed adjacent or parallel to the first track. In some embodiments, the collapsible cart may include a top cover with at least one integrated lock assembly. The at least one lock assembly removably couples the top cover to an interior surface of at least one of the front wall, the rear wall, the right sidewall, and the left sidewall.

[0009] In some embodiments, the left sidewall including a first left panel rotatably coupled to a second left panel. The second left panel may be proportioned to fit within an opening in the first

left panel. In some embodiments, the collapsible cart may include a second track formed along the first left panel and the second left panel extending from a first position on the first left panel to a second position on the second left panel. Embodiments may also include a second slideable member cooperatively engaged to the second track, the second slideable member may be movable along the second track between an open position to a closed position to selectively lock the first left panel to the second left panel.

[0010] In some embodiments, the second slideable member may be in the open position when disposed along the second track adjacent the first position of the second track while not disposed along the second left panel and may be in the closed position when disposed along the second track adjacent the second position of the second track while being disposed across both the first left panel and second left panel.

[0011] In yet another embodiment, the right sidewall further including a third right panel. In some embodiments, each of the second right panel and the third right panel encloses half of the opening in the first right panel. In some embodiments, the collapsible cart may include a telescoping handle assembly adjacent the rear wall, the telescoping handle assembly may include a hand grip at a distal end of the telescoping handle assembly and may be pivotably coupled at proximal end to the bottom of the rear wall. Embodiments may also include a sliding lock member translatable along at least a portion of the telescoping handle assembly. In some embodiments, the sliding lock member may be configured to lock to the rear wall securing the telescoping handle assembly parallel to the rear wall. In some embodiments, the sliding lock member may be configured to unlock from the rear wall allowing the telescoping handle assembly to pivot at an angle away from the rear wall. In some embodiments, the telescoping handle assembly may be rotatable to be horizontally aligned with the bottom wall and may be retained parallel to the bottom wall when in storage. In some embodiments, the bottom wall may include a sleeve. The telescoping handle assembly may be rotatable to be horizontally aligned with the bottom wall and may be retained in the sleeve when the stackable collapsible cart may be in the closed condition.

[0012] In some embodiments, the collapsible cart may include a wheel assembly coupled to the bottom wall of the cart. In some embodiments, the rigid top cover includes an indentation pattern being at least substantially aligned with the vertical axis of the wheel assembly. The indentation pattern configured to receive a wheel assembly from another identical collapsible cart when stacked vertically. In some embodiments, the top cover may also include an integrated second lock assembly. The second lock assembly may be configured to removably couple the top cover to an interior surface of at least one of the front wall, the rear wall, the right sidewall, and the left sidewall.

[0013] Each of the foregoing various aspects, together with those set forth in the claims and described in connection with the embodiments summarized above and disclosed herein may be combined to form claims for a device, apparatus, system, methods of manufacture and/or use in any way disclosed herein without limitation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] These and other features, aspects and advantages are described below with reference to the drawings, which are intended to illustrate but not to limit the invention. In the drawings, like reference characters denote corresponding features consistently throughout similar embodiments.

[0015] FIG. 1 is a perspective view of a collapsible cart illustrated in an open condition, according to an embodiment.

[0016] FIG. 2 is a right-side view of the collapsible cart of FIG. 1, according to an embodiment.

[0017] FIG. 3 is a left-side view of the collapsible cart of FIG. 1, according to an embodiment.

[0018] FIG. **4** is a front plan view of the collapsible cart of FIG. **1**, according to an embodiment.

[0019] FIG. **5** is a rear plan view of the collapsible cart of FIG. **1**, according to an embodiment.

[0020] FIG. **6** is a top view of the collapsible cart of FIG. **1**, according to an embodiment.

[0021] FIG. **7** is a bottom view of the collapsible cart of FIG. **1**, according to an embodiment.

[0022] FIG. **8** is a perspective view of the collapsible cart of FIG. **1** illustrated with the sidewalls folded inwardly, according to an embodiment.

[0023] FIG. **9** is a perspective view of the collapsible cart of FIG. **1** illustrated in a closed condition, according to an embodiment.

[0024] FIG. **10** is a perspective view of a collapsible cart illustrated in an open condition, according to an embodiment.

[0025] FIG. **11** is a right-side view of the collapsible cart of FIG. **10**, according to an embodiment.

[0026] FIG. **12** is a left-side view of the collapsible cart of FIG. **10**, according to an embodiment.

[0027] FIG. **13** is a front plan view of the collapsible cart of FIG. **10**, according to an embodiment.

[0028] FIG. **14** is a rear plan view of the collapsible cart of FIG. **10**, according to an embodiment.

[0029] FIG. **15** is a top view of the collapsible cart of FIG. **10**, according to an embodiment.

[0030] FIG. **16** is a bottom view of the collapsible cart of FIG. **10**, according to an embodiment.

[0031] FIG. **17** is a perspective view of the collapsible cart of FIG. **10** illustrated with the sidewalls folded inwardly, according to an embodiment.

[0032] FIG. **18** is a perspective view of the collapsible cart of FIG. **10** illustrated in a closed condition, according to an embodiment;

[0033] FIG. **19** is a right-side view of the collapsible cart similar to FIG. **1** illustrating another embodiment having the rigid cover panel pivoting in a first axis by a hinge mechanism that connects the cover panel to the front wall.

[0034] FIG. **20** is a view similar to FIG. **19** now showing the rigid cover panel in a vertical orientation.

[0035] FIG. **21** is a front plan view of the structure of FIG. **19**.

[0036] FIG. **22** is a view similar to FIG. **21** now showing the rigid cover panel rotating in a second axis.

[0037] FIG. **23** is a view similar to FIG. **21** and FIG. **22** now showing the rigid cover panel further rotating in a second axis.

[0038] FIG. **24** is a perspective view showing the magnets and washers disposed inside the collapsible cart.

[0039] FIG. **25** is a perspective view showing the hinge mechanism of FIGS. **20-24**. FIG. **26** is an enlarged view taken along line **26** of FIG. **25**.

[0040] FIG. **27** is a perspective view of another embodiment of the collapsible cart now having a pivoting telescoping handle.

[0041] FIG. **28** is a view similar to FIG. **27** now showing a locking slide being moved upwards to unlock the telescoping handle relative to the rear wall of the cart.

[0042] FIG. **29** is a perspective view of the structure of FIGS. **27-28** now showing how the telescoping handle can pivot at its proximal end for rolling the cart while allowing all four wheels to remain in contact with the ground.

[0043] FIG. **30** is a perspective view of a collapsible cart illustrated in a closed condition, according to an embodiment.

[0044] FIG. **31** is another perspective view of the collapsible cart of FIG. **30**, according to an embodiment.

[0045] FIG. **32** is a perspective view of the collapsible cart of FIG. **30** having a cover panel stored and engaged to the underside of the collapsible cart, according to an embodiment.

[0046] FIG. **33** is a view similar to FIG. **32** showing the cover panel removed from the underside of the collapsible cart of FIG. **30**, according to an embodiment.

[0047] FIGS. **34-41** are perspective views of the collapsible cart of FIG. **30** illustrating the

transition from a closed condition to an open condition, according to an embodiment.

[0048] FIG. **42A** is a perspective view of the collapsible cart of FIG. **30** illustrating a second right panel rotatably coupled to a first right panel, according to an embodiment.

[0049] FIG. **42B** is a cross-sectional view of a first slideable member for coupling the first right panel to the second right panel and a second slideable member for coupling the third right panel to the second right panel, according to an embodiment.

[0050] FIG. **43** is a perspective view of the collapsible cart of FIG. **30** illustrating a second right panel and a third right panel each rotatably coupled to a first right panel, according to an embodiment.

[0051] FIG. **44** is a perspective view of a collapsible cart with a lock mechanism illustrated in an open condition, according to an embodiment.

[0052] FIG. **45** is a perspective view of another collapsible cart illustrated in a closed condition and having a cover panel stored on the top of the collapsible cart, according to an embodiment.

[0053] FIG. **46** is a perspective view of the underside of the collapsible cart of FIG. **45**, according to an embodiment.

[0054] FIG. **47** is a close-up view of a wheel assembly engaged to a receiving aperture in the underside of the collapsible cart of FIG. **45**, according to an embodiment.

[0055] FIG. **48** is a perspective view of the underside of the collapsible cart of FIG. **45** with wheel assemblies, according to an embodiment.

[0056] FIG. **49** is a perspective view of the collapsible cart of FIG. **45** without the cover panel, according to an embodiment.

[0057] FIGS. **50-51** are perspective views of the collapsible cart of FIG. **45** illustrating the transition from a closed condition to an open condition, according to an embodiment.

[0058] FIGS. **52-53** illustrate a lock assembly translatable to lock a first panel to a second panel of the collapsible cart of FIG. **45**, according to an embodiment.

[0059] FIG. **54** is a perspective view of the collapsible cart of FIG. **45** in an open condition with the cover panel placed on top, according to an embodiment.

[0060] FIG. **55** illustrate a lock assembly translatable to lock the cover panel to one of the sidewalls of the collapsible cart of FIG. **45**, according to an embodiment.

[0061] FIG. **56** is a perspective view of the collapsible cart of FIG. **45** illustrating a second panel rotatably coupled to a first panel, according to an embodiment.

[0062] FIG. **57** is a perspective view of the collapsible cart of FIG. **45** in an open condition with alternate second panel having ribbed configuration for increased strength, according to an embodiment.

[0063] FIG. **58** is a view similar to FIG. **57** showing how the telescoping handle extended from underneath the collapsible cart and pivoted at its proximal end, according to an embodiment.

[0064] FIGS. **59-61** illustrate the locking of the telescoping handle to the cover panel so as to retain the telescoping handle in a vertical fixed position, according to an embodiment.

[0065] FIG. **62** illustrate the collapsible cart of FIG. **45** showing a locking slide being moved upwards to unlock the telescoping handle relative to the cover panel of the cart.

[0066] FIG. **63** is a view similar to FIG. **62** now showing how the telescoping handle can pivot at its proximal end for rolling the cart while allowing all four wheels to remain in contact with the ground.

DETAILED DESCRIPTION

[0067] Unique and inventive collapsible carts are disclosed herein. Although embodiments of collapsible carts are disclosed herein, it is to be expressly understood that the present invention is not restricted solely to such embodiments. Rather, the present disclosure is directed to each of the inventive features described below, both individually as well as collectively, in various embodiments. Further, as will become apparent to those skilled in the art, one or more aspects of the present disclosure may be incorporated in other devices.

[0068] FIGS. 1-9 illustrate an embodiment of a collapsible cart **10** with swivel wheels, according to an embodiment. The collapsible cart **10** may include a rigid frame **12** forming a compartment **14**. The rigid frame **12** may have a front wall **16**, a rear wall **18**, a right sidewall **20**, a left sidewall **22**, and a bottom wall **24**. The right sidewall **20** may include a first right panel **26** and a second right panel **28**. The first right panel **26** may be coupled with a first hinge **27** to the second right panel **28** along a first vertical axis **30**. The left sidewall **22** may include a first left panel **32** and a second left panel **34**. The first left panel **32** may be coupled with a second hinge **33** to the second left panel **34** along a second vertical axis **36**.

[0069] As can be appreciated, the first right panel **26**, the second right panel **28**, the first left panel **32** and the second left panel **34** may be formed of molded rigid plastic. In one embodiment, the first right panel **26**, the second right panel **28**, the first left panel **32** and the second left panel **34** may each be a solid panel member. In another embodiment, the first right panel **26**, the second right panel **28**, the first left panel **32** and the second left panel **34** may each be configured in a frame structure with a plurality of apertures **38** positioned between vertical and horizontal rigid members **39**. Artisans would appreciate that this configuration allows for light-weight construction of the sidewalls **20** and **22**.

[0070] In one embodiment, the collapsible cart **10** may include a rotatable base panel **40** (shown in FIGS. 6 and 8) rotatably coupled to the bottom wall **24** within the compartment **14**. The rotatable base panel **40** may include a lower surface **42** (shown FIG. 8) and an upper surface **44** (shown FIG. 6). As can be appreciated, when in use, the rotatable base panel **40** may be rotated so that its lower surface rests against an interior surface of the bottom wall **24**.

[0071] In another embodiment, the collapsible cart **10** may include a first track **46** and a second track **48**. The first track **46** may extend across the first vertical axis **30** from a first position **50** on the first right panel **26** to a second position **52** on the second right panel **28**. The second track **48** may extend across the second vertical axis **36** from a first position **54** on the first left panel **32** to a second position **56** on the second left panel **34**.

[0072] The collapsible cart **10** may also include a first slideable member **58** and a second slideable member **60**. The first slideable member **58** may cooperatively engage the first track **46** and may be movable between an open position to a closed position to selectively secure or lock the first right panel **26** to the second right panel **28** (shown in FIG. 2). As can be appreciated, the first slideable member **58** is in the open position when adjacent the first position **50** of the first track **46** and is in the closed position when adjacent the second position **52** of the first track **46**. Similarly, the second slideable member **60** may be cooperatively engage the second track **48** and may be movable between an open position to a closed position to selectively secure or lock the first left panel **32** to the second left panel **34**. As can be appreciated, the second slideable member **60** is in the open position when adjacent the first position **54** of the second track **48** and is in the closed position when adjacent the second position **56** of the second track **48**. In another embodiment, the collapsible cart **10** may include one only one slideable member on either one of the sidewalls **20** or **22**. Alternatively, the collapsible cart **10** may include a plurality of slideable members for each sidewall **20** and **22**.

[0073] In one embodiment, the collapsible cart **10** may include a rigid cover panel **62** and a retractable handle **64**. The rigid cover panel **62** may be formed of a molded plastic with a ribbed pattern for increased rigidity. The rigid cover panel **62** may be configured to conform in shape and fit snugly to a top opening **63** of the compartment **14**. The rigid cover panel **62** may securely fit in a first position over the top opening **63** to serve as a cover or seat on top of the collapsible cart **10**. The rigid cover panel **62** may include one or more protrusions on a bottom surface to removably secure to the front wall **16** of the collapsible cart **10**. Meanwhile, the retractable handle **64** may be positioned adjacent the back wall **12**.

[0074] As shown in FIGS. 1-9, the collapsible cart **10** may include a plurality of rotatable swivel wheels **66** coupled to the bottom wall **24** of the collapsible cart **10**. The rotatable swivel wheels **66**

may include at least one-wheel locking assembly **68** having a first condition for locking at least one of the rotatable swivel wheels to prevent rolling movement, and a second condition for unlocking the at least one of the rotatable swivel wheels. As can be appreciated, the at least one-wheel locking assembly may include a brake actuator pedal **70**.

[0075] FIGS. **10-18** illustrate an embodiment of a collapsible cart **100** with a pair of three-wheel assemblies **102**, according to an embodiment. The collapsible cart **100** may include a spindle **104** rotatably coupled to the bottom wall **24** and adjacent an intersection of the bottom wall **24** and the rear wall **18**. Each three-wheel assembly **102** may include three wheels **106**, three spokes **108** and a central rotational point **110**. As can be appreciated, each wheel **106** may be rotatably coupled to one of the three spokes **108**, and each spoke **108** may be connected to the central rotational point **110** coupled to the spindle **104**.

[0076] As can be appreciated, the collapsible cart **10** may be fabricated with a light-weight material, such as plastic. In one embodiment, one or more of the walls **16**, **18**, **20**, **22** and **24** may be formed of a molded plastic with a ribbed pattern for increased rigidity.

[0077] In operation, the collapsible cart **10** and **100** is transitioned from a closed condition where it is folded up to an open position where it is expanded for use. In the closed condition (shown in FIGS. **9** and **18**), a user may unlock clips **72** on the right side and left side to extend the right and left opposing sidewalls **20** and **22**. In one embodiment, the user may need to insert his or hands in the compartment **14** to push out the sidewalls **20** and **22**. As the sidewalls **20** and **22** are pushed outwards, the rotatable base panel **40** drops down such that its lower surface rests against the interior surface of the bottom wall **24**, and stabilizes the cart by fitting tightly between the lower region of the two opposing sidewalls **20** and **22**. The first slideable member **58** and the second slideable member **60** may then be moved along their respective tracks **46** and **48** and from their respective open position to their respective closed position to selectively secure the sidewall panels. As can be appreciated, the slideable members **58** and **60** may be used to hold the extended sidewalls **20** and **22** in place. The rigid cover panel **62**, which may be secured to the front wall **16** for storage, may be unlatched and securely fit over the top opening **63** of the compartment **14** to serve as a cover or seat. The collapsible cart **10** or **100** may be used to transport heavy loads in this open condition or may be collapsed in reverse fashion to place in a closed condition for storage.

[0078] FIGS. **19-29** illustrate another embodiment of the present invention. The collapsible cart **10** is very similar to the previous embodiments already disclosed, such that common reference numbers are used herein. Similarly, the rigid frame **12** forms the compartment **14**. The rigid frame **12** includes the front wall **16**, the rear wall **18**, the right sidewall **20**, the left sidewall **22**, and the base panel **40**. The retractable handle mechanism **64** is disposed at, within or adjacent the back wall **18** as best shown in FIG. **27**. The retractable handle mechanism **64** comprises a hand grip **120** attached to a telescoping assembly **122**. The hand grip is attached at a distal end **124** of the telescoping assembly. The telescoping assembly **122** is pivotably attached at a proximal end **126** to the bottom **18b** of the rear wall **18** as best seen in FIG. **28**. As can be appreciated, there are a multitude of mechanisms and methods that may be used by those skilled in the art to create the pivot joint at the proximal end of the telescoping assembly, as this teaching is not to be limited to any one specific structure.

[0079] The rigid cover panel **62** still conforms in shape to a top opening **63** of the compartment as the rigid cover panel is configured to securely fit in a first position over the top opening to serve as a cover and/or a seat as shown in FIG. **1**. However, as now shown in FIG. **19**, the rigid cover panel is movably attached to a top **16a** of the front wall. The rigid cover panel is configured to be moved to a second position residing adjacent the front wall outside the compartment as best shown in FIG. **23**.

[0080] FIG. **19** shows the rigid cover panel **62** starting to pivot upwards about a first axis **82** and then FIG. **20** shows the rigid cover panel in a vertical orientation. FIG. **21** is a front view of the FIG. **20**. The cover panel **62** is able to pivot upwards due to a novel hinge mechanism **80** which is

best shown in FIGS. **24**, **25** and **26**. FIG. **26** is a close up of the hinge mechanism taken from FIG. **25**.

[0081] FIG. **26** shows that the hinge mechanism can pivot in both the first axis **82** and also a second axis **84**. The first axis and second axis are perpendicular to one another. Accordingly, the cover panel first pivots about the first axis **82** in FIGS. **19-21**. Then, in FIG. **22**, the cover panel rotates about the second axis **84** and continues to rotate in FIG. **23**. As shown in FIG. **23**, the cover panel is positioned out of the way of the compartment such that it is easy to place and remove items from the compartment. Yet, the cover panel will no longer be lost or unavailable as it is permanently connected to the front wall **16**.

[0082] The hinge mechanism shown in FIG. **26** comprises a first portion **86** and a second portion **88**. The first portion is able to rotate about the second axis **84** in relationship to the second portion **88**. In this embodiment, the hinge mechanism comprises two parts, but could comprise one part or any number of parts to achieve the multi-axis movement and shown and described herein.

[0083] The first portion of the hinge at a proximal end **90** is pivotably connected to the top of the front wall. In this case, it is attached on the inside of the front wall but could be attached to the outside as well. In this embodiment, the proximal end **90** has cylindrical extensions **94** that are configured to snap into circular recesses **96** formed into the front wall. Thus, the cylindrical extensions are able to pivot within the circular recesses. It is understood that other devices or structures could achieve the same movements, such as an auxiliary pin used for an axle that would extend through both the front wall and the proximal end of the front part. Accordingly, this teaching is not to be limited to the precise form shown and described herein.

[0084] The second portion of the hinge is connected at a distal end **92** to the cover panel **62** through the use of clips **90** and fasteners **92**. As is understood by those skilled in the art, the second portion **88** could be connected to the cover panel **62** through any combination of clips and/or fasteners and/or bonding agents (not shown). Furthermore, the second portion **88** could be formed as part of the cover panel **62** when the cover panel is injection molded during production.

[0085] In another embodiment not shown, the hinge mechanism could be replaced with a fabric strap where the fabric strap is inherently flexible to allow the multi-axis movement required.

[0086] Once the cover panel is in the open position as shown in FIG. **26**, it may swing back and forth undesirably as the cart is moved around. Thus, the inventors have further improved the novel hinge mechanism with a cam and flexure mechanism. A raised bump **100** is formed as part of a cam **101** that is made as part of the first portion **86** or attached to the first portion **86**. The cam **101** is generally circular in shape except for the raised bump **100**. The cam is generally hidden from view as being disposed behind the second portion **88**, but the raised bump **100** can be seen through the hole of the flexure **102** formed in the second portion **88**. Here, two flexures **102a** and **102b** are formed, but one flexure could have been used. The flexures abut the raised bump and create an interference fit that helps hold the cover panel into location once the raised bump is disposed between the two flexures. To move the cover panel out of the position, the raised bump engages at least one flexure and causes it to deflect a small distance which in turn creates a resistance to movement and a resistance for swinging back and forth. As is understood by those skilled in the art, a number of different cam and flexure mechanisms are possible, as this teaching is not to be limited to the precise form described herein.

[0087] There is also another solution that can be added to the previous solution or used in lieu of the previous solution to prevent or reduce the cover panel from swinging undesirably back and forth. Referring now to FIG. **24**, magnets **104a** and **104b** can be disposed as part of the front wall **16**. Then, additional magnets or ferromagnetic material, such as metal washers **106**, can be attached to the inside surface of the cover panel **62**. When the cover panel is used to close the top of the cart, the magnets **104** and washers **106** are far apart. Yet, when the cover panel is opened and disposed as shown in FIG. **26**, they are appropriately positioned such that they align with one another. Therefore, a magnetic attraction occurs between the magnets and washers that keeps the cover

panel in its position relative to the cart and prevents or reduces the undesirable swinging.

[0088] It is shown in FIG. 24 that the washers **106** are held with fasteners **107**, but they could also be bonded into place or securely held through an interference fit. Likewise, the magnets **104** can be screwed, bonded or held in place with an interference fit. The magnets **104** could be placed on either the outside of the front wall or on the inside of the front wall. Here, the magnets are shown in the inside of the front wall.

[0089] The rigid cover panel may also include a locking mechanism **110** configured to lock or unlock the rigid cover panel in the first position. The locking mechanism can be a switch that can be slid forwards and backwards to engage the rear wall.

[0090] A locking slide **128** is translatable along at least a portion of the telescoping assembly **122**. The locking slide **128** is a rigid structure that captures both of the telescoping tubes and helps secure them in parallel relationship. It is understood by those skilled in the art that just one telescoping tube could be used and as such the locking slide **128** would translate along this one telescoping tube.

[0091] In FIG. 27 the locking slide **128** is locking the telescoping assembly **122** to the rear wall **18** at a distance apart from the proximal end which is pivotable. Then, in FIG. 28, the locking slide **128** has moved upwards and no longer locks the telescoping assembly **122** to the rear wall **18**.

[0092] The locking slide **128** may be configured to lock to the rear wall **18** securing the telescoping assembly **122** parallel to the rear wall **18**. The locking slide **128** may also be configured to unlock from the rear wall **18** allowing the telescoping assembly **122** to pivot at an angle away from the rear wall **18**, as shown in FIG. 28. A release button **130** may be used to lock and unlock the locking slide **128** from the rear wall **18**. As shown in FIG. 28, the collapsible cart **10** can now be rolled with all four wheels remaining on the ground while still retaining the functionality of the cart as described in the previous embodiments.

[0093] It is also understood by those skilled in the art that the locking slide **128** could be replaced with a locking mechanism that either secures the telescoping assembly **122** to the rear wall **18** or allows the telescoping assembly **122** to pivot about the proximal end. In other words, in other embodiments, the locking slide **128** need not translate along the telescoping assembly **122** while still being able to lock and unlock the telescoping assembly **122** in relation to the rear wall **18**.

[0094] FIGS. 30-44 illustrate another embodiment of the present invention. The collapsible cart **10** is very similar to the previous embodiments already disclosed, such that common reference numbers are used herein. Like the prior disclosed embodiments, the rigid frame **12** forms the compartment **14**. The rigid frame **12** includes the front wall **16**, the rear wall **18**, the right sidewall **20**, the left sidewall **22**, and the bottom wall **24**. In some embodiments, the right sidewall **20** and the left sidewall **22** may be configured to fold inwardly in the closed condition. The right sidewall **20** may include a first right panel **26** rotatably coupled to a second right panel **28**.

[0095] The collapsible cart **10** may include a first track **46** formed along the first right panel **26** and the second right panel **28** extending from a first position on the first right panel **26** to a second position on the second right panel **28**. The collapsible cart **10** may also include a first lock assembly **42A** comprising a first slideable member **58** traversing the first track **46** and capable of cooperatively engaging with a peg **42B**. In one embodiment, the first slideable member **58** is spring loaded to move the first slideable member **68** from an open condition to a closed condition to lock around the peg **42B**. The first slideable member **58** may be movable along the first track **46** between an open position to a closed position to selectively lock the first right panel **26** to the second right panel **28**. As can be appreciated, other lock assemblies may be employed to selectively lock and/or unlock the first right panel **26** to the second right panel **28**.

[0096] In one embodiment, the second right panel **28** may be proportioned to fit within an opening in the first right panel **26**. In some embodiments, the second right panel **28** may be configured to conform in shape to cover at least a portion of an opening in the first right panel **26**. The first slideable member **58** may be in the open position when disposed along the first track **46** adjacent

the first position of the first track **46** while not disposed along the second right panel **28** and may be in the closed position when disposed along the first track **46** adjacent the second position of the first track **46** while being disposed across both the first right panel **26** and second right panel **28**. [0097] As shown in FIG. **44**, in some embodiments, the second right panel **28** may also include a ribbed wall **29** with a plurality of ribs **29a**. The ribbed wall **29** may be disposed adjacent the first track **46** (not shown). In some embodiments, the plurality of ribs **29a** may be aligned parallel to the first track **46** to distribute compressive stresses away from the first track **46**. In other embodiments, the plurality of ribs **29a** may be aligned vertically to withstand compressive forces applied to the right sidewall **20** from the weight of other objects resting on the top cover **62**, such as other stackable collapsible carts **10**.

[0098] In some embodiments, the collapsible cart **10** may also include a top cover **62** with at least one integrated lock assembly **65**. The top cover **62** may be configured to conform in shape to a top opening of the compartment. In one embodiment, the top cover **62** securely fits in a first position over the top opening to serve as a cover or seat on top of the collapsible cart **10**. The at least one lock assembly **65** may be configured to removably couple the top cover **62** to an interior surface of at least one of the front wall **16**, the rear wall **18**, the right sidewall **20**, and the left sidewall **22**.

[0099] As shown in FIGS. **35-43**, in some embodiments, the right sidewall **20** may include a third right panel **31**. In such embodiment, the second right panel **28** and the third right panel **31** may be configured to enclose half of the opening in the first right panel **26**. In yet another embodiment, the second right panel **28** and the third right panel **31** may conform in shape to collectively cover the opening in the first right panel **26**.

[0100] In some embodiments, as shown in FIG. **46**, the telescoping handle assembly **64** may be rotatable to be horizontally aligned with the bottom wall **24** and may be retained parallel to the bottom wall **24** when in storage. As with prior embodiments, the collapsible cart **10** shown in FIG. **48** may include a wheel assembly **66** coupled to the bottom wall **24** of the cart **10**.

[0101] In some embodiments, as shown in FIG. **51**, the left sidewall **22** may also include a first left panel **32** rotatably coupled to a second left panel **34**. In yet another embodiment, the second left panel **34** may be proportioned to fit within an opening in the first left panel **32**. In some embodiments, the collapsible cart **10** may also include a second track **48** formed along the first left panel **32** and the second left panel **34** extending from a first position on the first left panel **32** to a second position on the second left panel **34**. The collapsible cart **10** may also include a second slideable member **60** cooperatively engaged to the second track **48**, the second slideable member **60** may be movable along the second track **48** between an open position to a closed position to selectively lock the first left panel **32** to the second left panel **34**.

[0102] In some embodiments, the second slideable member **60** may be in the open position when disposed along the second track **48** adjacent the first position of the second track **48** while not disposed along the second left panel **34** and may be in the closed position when disposed along the second track **48** adjacent the second position of the second track **48** while being disposed across both the first left panel **32** and second left panel **34**.

[0103] As shown in FIGS. **58-63**, in some embodiments, the collapsible cart **10** may include a telescoping handle assembly **64** adjacent the rear wall **18**. The telescoping handle assembly **64** may include a hand grip **120** at a distal end **124** of the telescoping handle assembly **64** and may be pivotably coupled at proximal end **126** to the bottom **18b** of the rear wall **18**. In one embodiment, the collapsible cart **10** includes a sliding lock member **128a** translatable along at least a portion of the telescoping handle assembly **64**. The sliding lock member **128a** may be configured to lock to the rear wall **18** securing the telescoping handle assembly **64** parallel to the rear wall **18**. The sliding lock member **128a** may be configured to unlock from the rear wall **18** and/or cover panel **62** allowing the telescoping handle assembly **64** to pivot at an angle away from the rear wall **18**. In some embodiments, the bottom wall **24** may include a sleeve **25**. The telescoping handle assembly **64** may be rotatable to be horizontally aligned with the bottom wall **24** and may be retained in the

sleeve 25 when the collapsible cart 10 is in the closed condition.

[0104] As shown in FIG. 62, in some embodiments, the top cover 62 may also include an indentation pattern 62a being at least substantially aligned with a vertical axis of the wheel assembly 66. The indentation pattern 62a may be configured to receive a wheel assembly 66 from another identical collapsible cart 10 when stacked vertically.

[0105] Although the various inventive aspects are herein disclosed in the context of certain preferred embodiments, implementations, and examples, it will be understood by those skilled in the art that the present invention extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. In addition, while a number of variations of the inventive aspects have been shown and described in detail, other modifications, which are within their scope will be readily apparent to those of skill in the art based upon this disclosure. It should be also understood that the scope this disclosure includes the various combinations or sub-combinations of the specific features and aspects of the embodiments disclosed herein, such that the various features, modes of implementation, and aspects of the disclosed subject matter may be combined with or substituted for one another. Thus, it is intended that the scope of the present invention herein disclosed should not be limited by the particular disclosed embodiments or implementations described above, but should be determined only by a fair reading of the claims.

[0106] Similarly, this disclosure is not be interpreted as reflecting an intention that any claim require more features than are expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following the Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment.

[0107] Further, all claim terms should be interpreted in their most expansive forms so as to afford the applicant the broadest coverage legally permissible. Although the embodiments have been described with reference to the drawings and specific examples, it will readily be appreciated by those skilled in the art that many modifications and adaptations of the processes, methods and apparatuses described herein are possible without departure from the spirit and scope of the embodiments as claimed herein. Thus, it is to be clearly understood that this description is made only by way of example and not as a limitation on the scope of the embodiments as claimed below.

Claims

1. A collapsible container configured to transition from a closed condition where it is folded up to an open condition where it is expanded for use, the collapsible container comprising: a frame defining a compartment, wherein: the frame comprises at least five walls, with at least three of the walls configured to rotatably fold inwardly when the container is folded up into the closed condition, two of which are opposing walls; the two opposing walls each being rotatably coupled to a wall that does not fold inwardly when the container is folded up into the closed condition; one of the two opposing walls consists of a first panel and a second panel; and the second panel rotatably coupled to the first panel; a first latch part disposed on an edge of the first panel when the collapsible container is in the closed condition and the open condition; and a second latch part disposed on an edge of the second panel when the collapsible container is in the closed condition and the open condition, wherein the first latch part and the second latch part are configured to mate with one another and hold the first and second panels in a common plane when the first and second latch parts are latched together; and when the first and second latch parts are unlatched from one another, the second panel is capable of rotating relative to the first panel.
2. The collapsible container of claim 1, wherein the two opposing walls each have a length, and two other walls each have a height which is less than half that length.

3. The collapsible container of claim 2, wherein the two opposing walls are configured to fold inwardly and downwardly to the closed condition.
4. The collapsible container of claim 3, wherein the two other walls are configured to fold inwardly and downwardly and are substantially coplanar with each other when folded to the closed condition.
5. The collapsible container of claim 1, 4, wherein the two other walls are configured to lay atop the two opposing walls when the collapsible container is folded in the closed condition.
6. The collapsible container of claim 5, further comprising a plurality of rotatable swivel wheels coupled to the fifth wall.
7. The collapsible container of claim 5, wherein when the first and second latch parts are unlatched from one another, the second panel is capable of rotating outwardly relative to the collapsible container.
8. The collapsible container of claim 5, wherein the second panel conforms in shape to cover an opening in the first panel.
9. A collapsible container configured to transition from a closed condition where it is folded up to an open condition where it is expanded for use, the collapsible container comprising: a frame defining a compartment, wherein: the frame comprises at least five walls, including a first wall, a second wall, a third wall, a fourth wall, and a fifth wall; at least three of the walls are configured to fold inwardly toward the fifth wall when the container is in the closed condition; the third wall comprises a first panel and a second panel, the second panel rotatably coupled to the first panel; a fastener configured to selectively secure the first panel and the second panel in a substantially coplanar alignment, the fastener comprising: a first fastener member integrally disposed on an edge of the first panel when the collapsible container is in the closed condition and the open condition; and a second fastener member integrally disposed on an edge of the second panel when the collapsible container is in the closed condition and the open condition; wherein the first fastener member and the second fastener member are configured to mate with one another to hold the first panel and the second panel in the substantially coplanar alignment; and wherein, when the first fastener member and the second fastener member are disengaged, the second panel is capable of rotating relative to the first panel.
10. The collapsible container of claim 9, wherein the first and second walls each have a height which is less than half the length of each of the third and fourth walls.
11. The collapsible container of claim 10, wherein the first and second walls are configured to fold inwardly and downwardly to the closed condition.
12. The collapsible container of claim 11, wherein the third and fourth walls are configured to fold inwardly and downwardly and are substantially coplanar with each other when folded to the closed condition.
13. The collapsible container of claim 12, wherein the third and fourth walls are configured to lay atop the first and second walls when the collapsible container is folded in the closed condition.
14. The collapsible container of claim 13, further comprising a plurality of rotatable swivel wheels coupled to the fifth wall.
15. The collapsible container of claim 13, wherein when the first and second fastener members are disengaged from one another, the second panel is capable of rotating outwardly relative to the collapsible container.
16. The collapsible container of claim 13, further comprising a top cover formed to fit securely in a first position over the top of the collapsible container in the open condition, and to fit securely to the collapsible container in a second position when the collapsible container is folded in the closed condition.
17. A collapsible container configured to transition from a closed condition where it is folded up to an open condition where it is expanded for use, the collapsible container comprising: a frame defining a compartment, wherein: the frame comprises a first wall, a second wall, a third wall, a

fourth wall, and a fifth wall; at least the first, second, and third walls are configured to fold inwardly toward the fifth wall to the closed condition; the third wall comprises a first panel and a second panel, the second panel rotatably coupled to the first panel; the first wall and the second wall each have a height from top to bottom; the third wall and the fourth wall each have a length from one edge to another; and the height of each of the first and second walls is less than half the length of each of the third and fourth walls, such that the first and second walls are substantially coplanar with each other when folded inwardly to the closed condition; a coupler configured to selectively secure the first panel and the second panel in a substantially coplanar alignment, the coupler comprising: a first coupler member disposed on an edge of the first panel when the collapsible container is in the closed condition and the open condition; and a second coupler member disposed on an edge of the second panel when the collapsible container is in the closed condition and the open condition; wherein the first coupler member and the second coupler member are configured to mate with one another to hold the first panel and the second panel in the substantially coplanar alignment; and wherein, when the first coupler member and the second coupler member are disengaged, the second panel is able to rotate relative to the first panel.

18. The collapsible container of claim 17, wherein the first, second, third, and fourth walls are configured to fold inwardly and downwardly to the closed condition.

19. The collapsible container of claim 18, wherein the first and second walls are configured to lay atop the third wall when the collapsible container is folded in the closed condition.

20. The collapsible container of claim 19, further comprising a top cover formed to fit securely in a first position over the top of the collapsible container expanded in the open condition, and to fit securely to the collapsible container in a second position when the collapsible container is folded in a closed condition.

21. The collapsible container of claim 17, wherein the second panel conforms in shape to cover an opening in the first panel.

22. The collapsible container of claim 17, wherein the first panel defines an opening, the third wall further comprises a third panel, and the second panel and third panel together cover the opening in the first panel.

23. The collapsible container of claim 19, wherein the third wall is configured to lay atop the fourth wall when the collapsible container is folded in the closed condition.

24. A collapsible container configured to transition from a closed condition where it is folded up to an open condition where it is expanded for use, the collapsible container comprising: a frame defining a compartment, wherein: the frame comprises at least five walls, including a first wall, a second wall, a third wall, a fourth wall, and a fifth wall; at least three of the walls are configured to fold inwardly toward the fifth wall when the container is in the closed condition; the third wall comprises a first panel and a second panel, the second panel rotatably coupled to the first panel; and a lock assembly integrally disposed on an edge of the first panel and on an edge of the second panel both when the collapsible container is in the closed condition and when the collapsible container is in the open condition, the lock assembly being configured to selectively secure the first panel and the second panel together in a substantially coplanar alignment, wherein: when the collapsible container is in the open condition and the first panel and the second panel are not selectively secured together in substantially coplanar alignment by the lock assembly, the second panel is capable of rotating relative to the first panel.

25. The collapsible container of claim 24, wherein the first, second, third, and fourth walls are configured to fold inwardly and downwardly to the closed condition.

26. The collapsible container of claim 25, wherein the first and second walls are configured to lay atop the third wall when the collapsible container is folded in the closed condition.

27. The collapsible container of claim 26, further comprising a top cover formed to fit securely in a first position over the top of the collapsible container expanded in the open condition, and to fit securely to the collapsible container in a second position when the collapsible container is folded in

a closed condition.

28. The collapsible container of claim 24, wherein the second panel conforms in shape to cover an opening in the first panel.

29. The collapsible container of claim 24, wherein the first panel defines an opening, the third wall further comprises a third panel, and the second panel and third panel together cover the opening in the first panel.

30. The collapsible container of claim 29, wherein the third wall is configured to lay atop the fourth wall when the collapsible container is folded in the closed condition.
